

Hongyi Yang

hongyiyangzju@gmail.com · github.com/hongyiyangzju-byte

RESEARCH INTERESTS

Integrated quantum photonics and on-chip quantum light sources; first-principles modelling of optoelectronic devices (NEGF, DFT/DFPT, Wannier functions); hybrid integration of low-dimensional materials on silicon photonic platforms.

EDUCATION

- **Zhejiang University**, Hangzhou, China (Sep 2023 - Jun 2027, expected)
 - B.Eng. in Optoelectronic Information Science and Engineering
 - *Chu Kochen Honors College* (Mixed Class, Honors Program)
 - **GPA:** 4.11/4.30
 - **Rank:** 4/162 (Chu Kochen Honors College), 4/102 (School of Optoelectronic Science and Engineering)

PUBLICATIONS

† equal contribution · * corresponding author

1. **Hongyi Yang**[†], Haipeng Zhu[†], Shiyu Feng, Zexi Lu, Xiyao Peng, Hongsheng Chen*, Sihan Zhao*, Dexin Ye*, and Haoliang Qian*. “Sub-Poissonian electroluminescence from resonant inelastic tunnelling in one-dimensional gold atomic chains.” *ACS Photonics*, 2026. Under revision; both referees recommend publication.
2. **Hongyi Yang**, Wendi Xia, Jun Li, Dexin Ye*, and Haoliang Qian*. “Physically Grounded Neural Networks: From Physical Substrates to Physics-Integrated Learning.” *Applied Physics Reviews*, 2026. Under review.

RESEARCH EXPERIENCE

Strain-Induced Ferroelectricity in FAPbI₃ (Jul 2026 - Oct 2026) *Xu Research Group, Stanford University, USA. Visiting Undergraduate Research Intern. Advisor: Prof. Sheng Xu*

- Built DFT models of epitaxially strained FAPbI₃ in Quantum ESPRESSO, with phonon calculations of the strained structures.
- Located the Pb atom off the centre of its iodine octahedron and mapped the double-well energy landscape along z, showing the barrier deepens as compressive strain increases.
- Related the polarization along z to the Pb off-centring, which follows it almost linearly, and identified the two wells as the origin of ferroelectric switching.

Electrically Driven Non-Classical Light from Gold Atomic Chains (Apr 2025 - Present) *State Key Laboratory of Extreme Photonics and Instrumentation, Zhejiang University. Advisor: Prof. Haoliang Qian*

- Built an end-to-end quantum-transport model of a gold atomic-chain junction: transport Hamiltonian from fully relativistic DFT (Quantum ESPRESSO) projected onto Wannier functions, vibrational modes from DFPT, leads coupled through bulk self-energies.
- Solved the non-equilibrium Green’s function problem without the wide-band approximation, treating electron-phonon coupling in the self-consistent Born approximation; implemented the pipeline in Python on HPC clusters.
- Computed current noise as well as current, showing the emitted electroluminescence is sub-Poissonian (Fano factor well below one) and quantifying how electron-phonon scattering and substitutional doping reshape the photon statistics.
- Coupled the transport model to full-wave simulations of a plasmonic nanocube dimer, linking the resonant inelastic channel and its surface-plasmon excitation to the emitted light.

Perovskite Optoelectronic Devices (Sep 2023 - Jun 2024) *International Research Center for Advanced Photonics, Zhejiang University. Advisor: Prof. Dawei Di*

- Learned the cleanroom process for perovskite devices, including spin-coating and encapsulation.
- Designed and built the optical setup used to measure their external quantum efficiency.

HONOURS & AWARDS

- **2025-2026:** Huawei Elite Scholarship
- **2025-2026:** First Prize, National Undergraduate Optoelectronic Design Competition
- **Oct 2025:** Top Ten Undergraduates, School of Optoelectronic Science and Engineering, Zhejiang University
- **2024-2025:** OPPO “Benfen” Scholarship (8 recipients nationwide per year)
- **2024:** First Prize, Zhejiang Provincial College Physics Competition
- **2023-2024:** National Scholarship, the highest national honour for undergraduates in China (top 0.2%)
- **2023-2024:** First-Class Scholarship of Zhejiang University; Excellent Student Title
- **2023-2025:** 3rd Place (Long Jump), 5th Place (Triple Jump), Zhejiang University Sports Meeting
- **2022:** First Prize, National High School Physics Olympiad (Hunan Province)

SKILLS

- **Programming:** Python, PyTorch, C/C++, LaTeX; high-performance computing (SLURM clusters)
- **Simulation:** custom NEGF transport codes, Quantum ESPRESSO (DFT/DFPT), Wannier90, COMSOL Multiphysics, Zemax
- **Hardware:** SolidWorks, Altium Designer
- **Languages:** Chinese (native); English, TOEFL 103 (R 27 / L 27 / S 25 / W 24)